**Lab Report-06**

Report on Introduction to C Programming

*Course Title: Structured Programming*

*Course Code:CSE - 103, Section-12*

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**Exercise 1:**

Write a C program to read elements in an array and print array. **Example:**

Input size: 10

Input elements: 1

2

3

4

5

6

7

8

9

10

Output: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Solution:

```
#include<stdio.h>


int main(){

int n;

scanf("%d",&n);

int a[n] ;

for(int i=0;i<10;i++){

    scanf("%d",&a[i]);
```

```
}

for(int i=0;i<10;i++){

    printf("%d",a[i]);

      if(i<9) printf(“,”);

}

return 0;
}
```

**Exercise 2**

Write a C program to read elements in an array and find the sum of all elements of the array.

**Example:**

Input elements: 10, 20, 30, 40, 50

Sum of all elements = 150

Solution:

**Solution:**

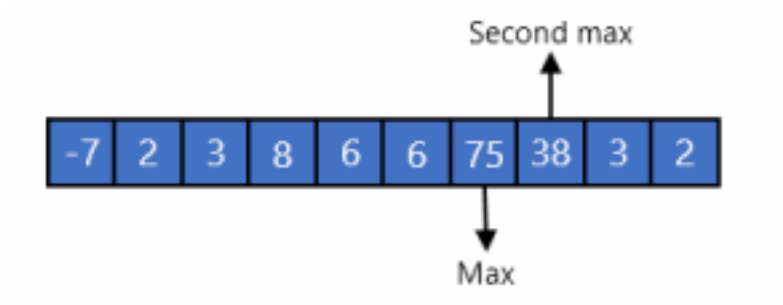
```
#include<stdio.h>

int main(){
int n;
scanf("%d",&n);
int a[n] ;
for(int i=0;i<n;i++){
    scanf("%d",&a[i]);
}
int sum =0;
for(int i=0;i<n;i++){
    sum+=a[i];
}
```

```
printf("Sum of all elements = %d",sum);
return 0;
}
```

## Exercise 3

Write a C program to find the largest and second largest element in an array.




```
Solution:
#include<stdio.h>

int main(){
int n;
scanf("%d",&n);
int a[n] ;
for(int i=0;i<n;i++){
    scanf("%d",&a[i]);
}
int largest = a[0];
for(int i=1;i<n;i++){
    if(a[i]>largest)
        largest = a[i];
}

int sec = -1000;
for(int i=0;i<n;i++){
    if(a[i]>sec && a[i]!=largest)
        sec = a[i];
}
printf("largest: %d\nSecond largest: %d",largest,sec);
return 0;
}
```

**Exercise 4**

Write a C program to insert an element in array at specified position. **Example:**

If the elements of array are: 10, 20, 30, 40, and 50

After inserting 25 at position 3

Elements of array are: 10, 20, 25, 30, 40, and 50

Solution:

```
#include<stdio.h>


int main(){

int n;

scanf("%d",&n);

int a[n+1] ;

for(int i=0;i<n;i++){

    scanf("%d",&a[i]);

}

int value, idx;

printf("Enter the position and value:\n");

scanf("%d%d",&idx,&value);

for(int i=n;i>idx;i--){
```

```
    a[i] = a[i-1];

}

a[idx] = value;

for(int i=0;i<=n;i++){

    printf("%d ",a[i]);

}

return 0;
}
```

**Exercise 5**

Write a C program to delete an element from an array at specified position. How to remove an element from array at given position.

**Example:**

Input elements:

10
20
30
40
50

Input position to delete: 2

Output: 10, 30, 40, 50

Solution:

```
#include<stdio.h>
```

```
int main(){

int n;

scanf("%d",&n);

int a[n] ;

for(int i=0;i<n;i++){

    scanf("%d",&a[i]);

}

int idx;

printf("Enter the position :\n");

scanf("%d",&idx);

for(int i=idx;i<n-1;i++){

    a[i] = a[i+1];

}

for(int i=0;i<n-1;i++){

    printf("%d ",a[i]);

}

return 0;
}
```

## Exercise 6

Write a C program to read elements in an array and find its reverse.

**Example:**
If the elements of the array are: 10, 5, 16, 35, 500
Then its reverse would be: 500, 35, 16, 5, 10

Solution:

```
#include<stdio.h>

int main(){
int n;
scanf("%d",&n);
int a[n] ;
for(int i=0;i<n;i++){
    scanf("%d",&a[i]);
}
printf("The reverse would be: ");
for(int i=n-1;i>=0;i--){
    printf("%d ",a[i]);
}
return 0;
}
```

**Exercise 7**

Write a program in C to count the total number of duplicate digits in a number using an array.
Test Data :
Input the number: 3523423

Expected Output :
Total number of duplicate elements found in the array is : 2

Solution:

```
#include<stdio.h>

int main(){
int n;
scanf("%d",&n);
int digit[10] = {0};
int t =n;
while(t){
    digit[t%10]++;
    t=t/10;
}
int cnt=0;
for(int i=0;i<10;i++){
       if(digit[i]>1){
        cnt++;
    }
}
printf("total number of duplicate elements in %d is %d",n,cnt);
return 0;
```

```
}
```

**Exercise 8**

Write a program in C to print all unique digits in a number using an array. Test Data :

Input the number: 35234237

Expected Output :
The unique elements found in the array are:
4 5 7

Solution:

```
#include<stdio.h>

int main(){
int n;
scanf("%d",&n);
int digit[10] = {0};
int t =n;
while(t){
    digit[t%10]++;
    t=t/10;
}
int cnt=0;
for(int i=0;i<10;i++){
       if(digit[i]==1){
        cnt++;
    }
}
printf("total number of unique digits in %d is %d",n,cnt);
return 0;
return 0;
}
```

**Exercise 9:**

Write a program in C for a 2D array of size 3x3 and print the matrix. Test Data:
Input elements in the matrix:
Element - [0],[0] : 1
Element - [0],[1] : 2
Element - [0],[2] : 3
Element - [1],[0] : 4
Element - [1],[1] : 5
Element - [1],[2] : 6

Element - [2],[0] : 7
Element - [2],[1] : 8
Element - [2],[2] : 9

Expected Output:
The matrix is:
1 2 3
4 5 6
7 8 9

Solution:

```
#include<stdio.h>

int main(){
int a[3][3];
for(int i=0;i<3;i++){
    for(int j=0;j<3;j++){
        scanf("%d",&a[i][j]);
    }
}
for(int i=0;i<3;i++){
    for(int j=0;j<3;j++){
        printf("%d ",a[i][j]);
    }
    printf("\n");
}
return 0;
}
```

**Exercise 10:**

Write a program in C for adding two matrices of the same size. Test Data:
Input the size of the square matrix (less than 5): 2 Input elements in the first matrix:
Element - [0],[0] : 1
Element - [0],[1] : 2
Element - [1],[0] : 3
Element - [1],[1] : 4
Input elements in the second matrix:
Element - [0],[0] : 5
Element - [0],[1] : 6
Element - [1],[0] : 7
Element - [1],[1] : 8

Expected Output:

The First matrix is:
1 2
3 4

The Second matrix is:
5 6
7 8

The Addition of two matrix is:
6 8
10 12

Solution:

```
#include<stdio.h>

int main(){
int r;
printf("Enter the number of row \n");
scanf("%d",&r);
int a[r][r];
for(int i=0;i<r;i++){
    for(int j=0;j<r;j++){
        scanf("%d",&a[i][j]);
    }
}
int b[r][r];
for(int i=0;i<r;i++){
    for(int j=0;j<r;j++){
        scanf("%d",&b[i][j]);
    }
}
int sum[r][r];
for(int i=0;i<r;i++){
    for(int j=0;j<r;j++){
       sum[i][j] = a[i][j] + b[i][j];
    }
}
printf("Summation Matrix :\n");
for(int i=0;i<r;i++){
    for(int j=0;j<r;j++){
        printf("%d ",sum[i][j]);
    }
    printf("\n");
}
return 0;
}
```

**Exercise 11:**

Write a program in C for the multiplication of two square matrices. Test Data:
Input the rows and columns of first matrix: 2 2 Input the rows and columns of second matrix: 2 2 Input elements in the first matrix:
Element - [0],[0] : 1
Element - [0],[1] : 2
Element - [1],[0] : 3
Element - [1],[1] : 4
Input elements in the second matrix:
Element - [0],[0] : 5
Element - [0],[1] : 6
Element - [1],[0] : 7
Element - [1],[1] : 8
Expected Output:

The First matrix is:
1 2
3 4
The Second matrix is:
5 6
7 8
The multiplication of two matrix is:

19 22
43 50

Solution:

```
#include<stdio.h>

int main(){
int r,c;
printf("Enter the number of row and column of matrix 1\n");
scanf("%d%d",&r,&c);
int a[r][c];
for(int i=0;i<r;i++){
    for(int j=0;j<c;j++){
        scanf("%d",&a[i][j]);
    }
}
int rr, cc;
printf("Enter the number of row and column of matrix 1\n");
scanf("%d%d",&rr,&cc);
int b[rr][cc];
for(int i=0;i<rr;i++){
    for(int j=0;j<cc;j++){
```

```
        scanf("%d",&b[i][j]);
    }
}
int product[cc][r];
for(int i=0;i<cc;i++){
    for(int j=0;j<r;j++){
        product[i][j] =0;
    }
}
for(int i=0;i<cc;i++){
        for(int k=0;k<r;k++){
                    for(int j=0;j<r;j++){
                    product[i][k] += a[i][j] * b[j][k];
                }
        }

}
printf("product Matrix :\n");
for(int i=0;i<cc;i++){
    for(int j=0;j<r;j++){
        printf("%d ",product[i][j]);
    }
    printf("\n");
}
return 0;
}
```

**Exercise 12:**

Write a program in C to find the transpose of a given matrix. Test Data:
Input the rows and columns of the matrix: 2 2 Input elements in the first matrix:
Element - [0],[0] : 1
Element - [0],[1] : 2
Element - [1],[0] : 3
Element - [1],[1] : 4
Expected Output:
The matrix is:
1 2
3 4

The transpose of a matrix is:
1 3
2 4

Solution:

```
#include<stdio.h>
```

```
int main(){
int r,c;
printf("Enter the number of row and column \n");
scanf("%d%d",&r,&c);
int a[r][c];
int t[c][r];
for(int i=0;i<r;i++){
    for(int j=0;j<c;j++){
        scanf("%d",&a[i][j]);
       t[j][i] = a[i][j];
    }
}
printf("The Matrix: \n");
for(int i=0;i<r;i++){
    for(int j=0;j<c;j++){
        printf("%d ",a[i][j]);
    }
    printf("\n");
}
printf("Transpose :\n");
for(int i=0;i<c;i++){
    for(int j=0;j<r;j++){
        printf("%d ",t[i][j]);
    }
    printf("\n");
}

return 0;
}
```

**Exercise 13:**

Write a program in C to find the sum of the right diagonals of a matrix. Test Data:
Input the size of the square matrix: 2
Input elements in the first matrix:
Element - [0],[0] : 1
Element - [0],[1] : 2
Element - [1],[0] : 3
Element - [1],[1] : 4
Expected Output:
The matrix is:
1 2
3 4
Addition of the right Diagonal elements is: 5

Solution:

```
#include<stdio.h>

int main(){
int r;
printf("Enter the number of row of the square matrix:  \n");
scanf("%d",&r);
int a[r][r];
for(int i=0;i<r;i++){
    for(int j=0;j<r;j++){
        scanf("%d",&a[i][j]);
    }
}
printf("The Matrix: \n");
for(int i=0;i<r;i++){
    for(int j=0;j<r;j++){
        printf("%d ",a[i][j]);
    }
    printf("\n");
}
int sum =0;
for(int i=0;i<r;i++){
    for(int j=0;j<r;j++){
        if(i==j){
            sum += a[i][j];
        }
    }
}
printf("Sum of right diagonals :%d",sum);


return 0;
}
```

**Exercise 14:**

Write a program in C to check whether a given matrix is an identity matrix. Test Data:
Input number of Rows for the matrix: 3
Input number of Columns for the matrix: 3 Input elements in the
first matrix:
Element - [0],[0] : 1
Element - [0],[1] : 0
Element - [0],[2] : 0
Element - [1],[0] : 0
Element - [1],[1] : 1
Element - [1],[2] : 0
Element - [2],[0] : 0
Element - [2],[1] : 0

Element - [2],[2] : 1
Expected Output:
The matrix is:
1 0 0
0 1 0
0 0 1
The matrix is an identity matrix.

Solution:

```
#include<stdio.h>

int main(){
int r;
printf("Enter the number of row of the square matrix:  \n");
scanf("%d",&r);
int a[r][r];
for(int i=0;i<r;i++){
    for(int j=0;j<r;j++){
        scanf("%d",&a[i][j]);
    }
}
printf("The Matrix: \n");
for(int i=0;i<r;i++){
    for(int j=0;j<r;j++){
        printf("%d ",a[i][j]);
    }
    printf("\n");
}
int flag =1;
for(int i=0;i<r;i++){
    for(int j=0;j<r;j++){
        if(i==j){
            if(a[i][j]!=1){
                flag =0;
                break;
            }
        }
        else{
            if(a[i][j] !=0){
                flag =0;
                break;
            }
        }
    }
    if(flag ==0){
        break;
    }
}
```

```
if(flag ==0){
    printf("Not Identity");
}
else{
    printf("Identity Matrix");
}

return 0;
}
```

**Exercise 15:**

Write a program in C to search for an element in a 2D matrix. Expected Output:

The given array in matrix form is:
15 23 31 39
18 26 36 43
25 28 37 48
30 34 39 50
The given value for searching is: 37
The element Found at the position in the matrix is: 2, 2

Solution:

```
#include<stdio.h>

int main(){
int r,c;
printf("Enter the number of row and column of the matrix:  \n");
scanf("%d%d",&r,&c);
int a[r][r];
for(int i=0;i<r;i++){
    for(int j=0;j<c;j++){
        scanf("%d",&a[i][j]);
    }
}
printf("The Matrix: \n");
for(int i=0;i<r;i++){
    for(int j=0;j<c;j++){
        printf("%d ",a[i][j]);
    }
    printf("\n");
}
printf("Enter the value to search:\n");
int value; scanf("%d",&value);
int flag =0;
int rr,cc;
for(int i=0;i<r;i++){
```

```
    for(int j=0;j<c;j++){
        if(value ==a[i][j]){
            flag =1;
            rr = i; cc=j;
            break;
        }
    }
    if(flag ==1){
        break;
    }
}

if(flag==1){
    printf("Value found\n at position %d,%d",rr,cc);
}
else{
    printf("Value Not found");
}


return 0;
}
```